

In a 50 kVA transformer, the iron loss = 500 W and full load copper loss is 800 W. Find the efficiency at full load and half load at 0.8 power factor lagging.?

For full load: $x = 1$.

$$\eta = \frac{x \times \text{full load in kVA} \times \text{pf}}{(x \times \text{full load kVA} \times \text{pf}) + (P_i + x^2 P_c)}$$

$$= \frac{1 \times 50 \times 0.8}{(50 \times 0.8) + (500 + 800 \times 10^3)}$$

$$= \underline{\underline{0.968}} \quad = \underline{\underline{0.968}}$$

$$\text{Efficiency} = 96.8\%$$

$$x = 0.5$$

$$\mu = 0.5 \times 50 \times 0.8$$

$$\frac{(\cancel{0.5} \times \cancel{50} \times 0.8) + 0.5 + (0.5)^2 0.8}{}$$

$$= \frac{20}{\cancel{20} + 0.5 + 0.2}$$

$$= \frac{20}{20.7} = \underline{\underline{0.966}}$$